

Fact-Checkers: How to Know What's Real

GRADES 3-4

30 MINUTES

GENERAL/HOMEROOM

Students learn that optical illusions show us our eyes can be fooled, and will develop strategies for verifying information before believing or sharing it.

LEARNING OBJECTIVE

Students will understand that optical illusions demonstrate how easily our eyes and brains can be deceived, and will develop and apply five practical strategies for verifying information before believing or sharing it.

MATERIALS NEEDED

- Color illusion printouts (checkerboard shadow illusion or identical colors in different contexts)
- Motion illusion printouts (rotating snakes or peripheral drift illusion)
- Teacher-created fake social media post example (age-appropriate, obviously fictional)
- Chart paper and markers for creating class "Fact-Checker Rules"
- Optional: Student handouts (three levels available)

LESSON PLAN

Hook: "Trust Your Eyes?" Challenge (5 minutes)

Opening demonstration: Show the checkerboard shadow illusion (where two squares appear different colors but are identical).

"I'm going to show you two squares. Tell me: Are they the same color or different colors?" Take a vote by show of hands. Reveal that they're identical: "These squares are EXACTLY the same color! Your brain changed what you saw based on shadows around them."

Second demonstration: Show a motion illusion (rotating snakes or similar). "Is this picture moving?" (It's completely still, but appears to move.)

"Your eyes are sending your brain information that isn't true. This is fun with pictures—but what happens when someone TRIES to trick you on purpose?"

Introduction: From Optical Illusions to Real Deception (5 minutes)

"Optical illusions are fun because we know they're tricks. But throughout history, people have used visual tricks to fool others on purpose."

Historical example: Briefly explain spirit photography (1860s-1900s): "Over 100 years ago, photographers discovered they could make 'ghosts' appear in pictures by double-exposing film. They charged grieving families money to 'photograph' their dead relatives."

Modern connection: "Today, we have 'deepfakes'—videos made by computers that make it look like someone said or did something they never actually did. And fake pictures and posts spread online every day."

"The good news? Just like you learned the shadow illusion was a trick, you can learn to spot fakes too. Today, you're becoming Fact-Checkers!"

Activity: Creating Our Fact-Checker Rules (12 minutes)

Part 1: Examining a Fake Post (4 minutes)

Show your teacher-created fake social media post. Example: A post claiming "Scientists discover chocolate is healthier than vegetables!" with a fake news source.

"Before I tell you if this is real or fake, what questions should we ask about it?"

Guide students to generate questions: Who said this? Where did it come from? Does this seem too good to be true?

Lesson Plan (continued)

Part 2: Building Our 5 Fact-Checker Rules (8 minutes)

On chart paper, create the class "Fact-Checker Rules" together. Guide students toward these five strategies:

The 5 Fact-Checker Rules

1. **STOP** before you share — Take a breath. Don't pass along information just because it's exciting or upsetting.
2. **CHECK** the source — Who created this? Is it from a real, trusted place?
3. **SEARCH** for more — Can you find this same information from other sources? If only one website says it, be suspicious.
4. **ASK** an adult — When in doubt, check with a parent, teacher, or librarian before believing or sharing.
5. **TRUST** your gut — If something seems too amazing, too scary, or too perfect—it might be fake.

Practice: Applying the Rules (5 minutes)

Give 2-3 quick scenarios and have students identify which Fact-Checker Rule(s) apply:

- "Your friend shows you a video of a celebrity saying something mean. What rules should you use?" (CHECK the source, SEARCH for more)
- "You see a post saying school is canceled tomorrow, but it's not from the school website." (CHECK the source, ASK an adult)
- "A website says you won a free video game, just click here!" (TRUST your gut, ASK an adult)

Closing: Connection to the Mini Museum (3 minutes)

"When you visit the Mini Museum, you'll see real artifacts that tricked people's eyes over 100 years ago: a stereoscope from 1904 that makes flat pictures look 3D, and illusion cards that transform when you move them. Remember: Understanding HOW tricks work makes you harder to fool!"

Mini Museum Connection

Option 1 — Visit the Mini Museum: The exhibition features a 1904 Monarch Stereoscope with original Keystone view cards. Victorian audiences were amazed by these devices just as we're amazed (and sometimes fooled) by digital manipulation today.

Option 2 — Virtual Visit: Explore at minimuseumproject.com/exhibitions/seeing-is-deceiving or view artifacts at minimuseumproject.com/exhibitions/seeing-is-deceiving/artifacts

ASSESSMENT

Formative: During practice scenarios, observe which students can correctly identify appropriate Fact-Checker Rules.

Exit ticket: "Write down the 5 Fact-Checker Rules in your own words, OR draw a picture showing one time you'll use them this week."

Extension: Students bring in examples of posts or claims they've seen that they think might be questionable.

DIFFERENTIATED STUDENT HANDOUTS

Three levels of student handouts are provided as separate files:

- **Green Level:** "More support" — Checklist approach, visual guides, fill-in-the-blank
- **Yellow Level:** "Ready for a challenge" — Apply rules to scenarios, independent work
- **Red Level:** "Go deeper" — Create own fact-check guide, analyze psychology of deception